

Lesson 5 Quantifiers

2 types :

① Universal quantifiers

For all, every, any, each, ... set

$\forall$ "for all"

Universal statement : $\forall x \in A$, $\underbrace{s(x)}_{\text{statement about } x}$

Ex For all positive integers n , $\underbrace{n^2 \geq n}_{s(n)}$
 $\forall n \in \mathbb{N}$

True : $\underbrace{n^2}_{n \cdot n} = n \cdot n \geq \underbrace{n \cdot 1}_n$

② Existential quantifiers

For some, there is/are, there exist(s), ...
one or more $\rightarrow$ at least one

Ex $\rightarrow$ For some natural numbers x , $\underbrace{2^x < x^2}_{s(x)}$
 $\exists$ Existence statement : $\exists x \in A, s(x)$

$\exists x \in \mathbb{N}$

There is a natural # x , such that $2^x < x^2$

True : $x = 3$ $\underbrace{2^3}_{8} < \underbrace{3^2}_{9}$

The sum of [✓] two even integers is even.

For all ^{any} integers x and y , if x, y are even,
then $x+y$ is even.

universal $\leftrightarrow$ conditional
if... then

A natural number is an integer.
^{Every} ----- $\nwarrow$ universal

There is a natural # which is an integer.
=...

$$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, x+y=0$$

For every integer x there is an integer y , s.t. $x+y=0$. $\nwarrow$ integer

True: $x + (-x) = 0$

$$\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, x+y=0$$

There is an integer x such that for every integer y , $x+y=0$

False: $\left. \begin{array}{l} x+0=0 \\ x+1=0 \end{array} \right\} \begin{array}{l} \rightarrow x=0 \\ \rightarrow x=-1 \end{array}$ ~~$\in \mathbb{Z}$~~

$$x + 2 = 0$$

⋮
1

Existence , Uniqueness

There is at least one

There is at most one

Unique existence : There is exactly one

Division algorithm (divide a by b)

$$a \in \mathbb{Z}, b \in \mathbb{N}$$

There are unique integers q (quotient) and r (remainder) s.t.

$$a = bq + r, \quad 0 \leq r < b$$

$$9 = 4(2) + 1$$

$$\begin{array}{r} \textcircled{2} \leftarrow q \\ 4 \overline{) 9} \\ \underline{8} \end{array}$$

$$\begin{array}{r} 1 \leftarrow q \\ \textcircled{4} \overline{) 9} \\ \underline{4} \\ \textcircled{5} \leftarrow r \end{array}$$

~~1 ← r~~

Negation

$$\sim(\forall x \in A, \underline{s(x)}) = \underline{\exists x \in A, \sim s(x)}$$

Example: For all integers x , $x^2 \geq x$

Neg: There is an integer x s.t. $x^2 < x$

$$\sim(\underline{\exists x \in A, s(x)}) = \underline{\forall x \in A, \sim s(x)}$$

Ex There is a natural # x such that $2^x < x^2$.

Neg: For all natural #'s x , $2^x \geq x^2$.

$$\begin{aligned} & \sim(\forall x \in \mathbb{Z}, \underline{\exists y \in \mathbb{Z}, x+y=0}) \\ &= \underline{\exists x \in \mathbb{Z}, \sim(\underline{\exists y \in \mathbb{Z}, x+y=0})} \\ &= \exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, \sim(\underline{x+y=0}) \\ &= \underline{\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, x+y \neq 0} \end{aligned}$$

There is an integer x such that for all integers y , $x+y \neq 0$.